

Fuwen TAN

CONTACT	fuwen.tan@gmail.com	https://fwtan.github.io/
SUMMARY	I'm a researcher in Efficient Machine Learning, working on making image/video generation models and LLMs faster. I am a main developer of MAI-Image-2.5-Flash and received a Best Paper Finalist at CVPR 2019.	
PUBLICATION	Progressive Mixed-Precision Decoding for Efficient LLM Inference Hao Mark Chen, Fuwen Tan, Alexandros Kouris, Royson Lee, Hongxiang Fan, Stylianos I. Venieris International Conference on Learning Representations (ICLR), 2025 MobileQuant: Mobile-friendly Quantization for On-device Language Models Fuwen Tan, Royson Lee, Lukasz Dudziak, Shell Xu Hu, Sourav Bhattacharya, Timothy Hospedales, Georgios Tzimiropoulos, Brais Martinez Conf. on Empirical Methods in Natural Language Processing, EMNLP Findings, 2024 Effective Self-supervised Pre-training on Low-compute Networks without Distillation Fuwen Tan, Fatemeh Saleh, Brais Martinez International Conference on Learning Representations (ICLR), 2023 EdgeViTs: Competing Light-weight CNNs on Mobile Devices with Vision Transformers Junting Pan, Adrian Bulat, Fuwen Tan, Xiatian Zhu, Lukasz Dudziak, Hongsheng Li, Georgios Tzimiropoulos, Brais Martinez European Conference on Computer Vision (ECCV), 2022 Instance-level Image Retrieval using Reranking Transformers Fuwen Tan, Jiangbo Yuan, Vicente Ordonez International Conference on Computer Vision (ICCV), 2021 Curriculum Labeling: Self-paced Pseudo-Labeling for Semi-Supervised Learning Paola Cascante-Bonilla, Fuwen Tan, Yanjun Qi, Vicente Ordonez AAAI Conference on Artificial Intelligence. (AAAI), 2021 Drill-down: Interactive Retrieval of Complex Scenes using Natural Language Queries Fuwen Tan, Paola Cascante-Bonilla, Xiaoxiao Guo, Hui Wu, Song Feng, Vicente Ordonez Conf. on Neural Information Processing Systems (NeurIPS), 2019 Text2Scene: Generating Compositional Scenes from Textual Descriptions Fuwen Tan, Song Feng, Vicente Ordonez Conf. on Computer Vision and Pattern Recognition (CVPR), 2019, Oral presentation Best Paper Finalist Where and Who? Automatic Semantic-Aware Person Composition Fuwen Tan, Crispin Bernier, Benjamin Cohen, Vicente Ordonez, Connelly Barnes Winter Conference on Applications of Computer Vision (WACV), 2018 FaceCollage: A Rapidly Deployable System for Real-time Head Reconstruction for On-The-Go 3D Telepresence Fuwen Tan, Chi-Wing Fu, Teng Deng, Jianfei Cai, Tat Jen Cham ACM Multimedia (ACM MM, full paper), 2017	

High-Quality Kinect Depth Filtering For Real-time 3D Telepresence
Mengyao Zhao, **Fuwen Tan**, Chi-Wing Fu, Chi-Keung Tang, Jianfei Cai, Tat Jen Cham
Conf. on Multimedia and Expo (**ICME**), 2013

Field-Guided Registration for Feature-Conforming Shape Composition
Hui Huang, Minglun Gong, Daniel Cohen-Or, Yaobin Ouyang, **Fuwen Tan**, Hao Zhang
SIGGRAPH Asia, 2012

EDUCATION	University of Virginia Ph.D. in Computer Science	Charlottesville, United States Aug.2015 - May.2021
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EXPERIENCE	Microsoft AI, United States <i>Member of Technical Staff, MAI Superintelligence</i>	Mar., 2026 - Present
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	Bytedance, United States <i>R&D in Efficient Machine Learning</i>	Jan., 2025 - Jan., 2026
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	Samsung AI Center, Cambridge, United Kingdom <i>R&D in Efficient Machine Learning</i>	Jun., 2021 - Jan., 2025
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TALKS	Toward Efficient On-device AI: the Model, the Training, and the Deployment <i>University of Surrey, November 2024</i>	
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SERVICE	Program Committee Siggraph Asia 2024 (Technical Communications & Posters)	
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Reviewer
Computer Vision: CVPR ([Outstanding Reviewer](#)), ECCV([Outstanding Reviewer](#)), ICCV, PAMI
Machine Learning: ICML([Outstanding Reviewer](#)), NeurIPS, ICLR, AAAI, IJCAI
Natural Language Processing: EMNLP

AWARDS	Computer Vision and Pattern Recognition (CVPR) Best Paper Finalist	2019
	European Conference on Computer Vision (ECCV) Outstanding Reviewer	2020
	Computer Vision and Pattern Recognition (CVPR) Outstanding Reviewer	2021
	International Conference on Machine Learning (ICML) Outstanding Reviewer	2022